

Distillation and Edge-Aware Loss for Efficient Monocular Depth Estimation

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Abstract

This paper studies efficient monocular depth estimation through knowledge distillation and edge-aware supervision. (placeholder stub)

1. Introduction

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3.1. Baseline

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4.1. Datasets

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Author Contributions.

References

- [1] Shariq Farooq Bhat, Reiner Birkel, Diana Wofk, Peter Wonka, and Matthias Müller. Zoedepth: Zero-shot transfer by combining relative and metric depth. *arXiv preprint arXiv:2302.12288*, 2023.
- [2] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*, 2021.
- [3] David Eigen, Christian Puhrsch, and Rob Fergus. Depth map prediction from a single image using a multi-scale deep network. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 27, 2014.
- [4] Andreas Geiger, Philip Lenz, Christoph Stiller, and Raquel Urtasun. Vision meets robotics: The kitti dataset. *The International Journal of Robotics Research (IJRR)*, 32(11):1231–1237, 2013.
- [5] Xiankang He, Dongyan Liu, Hao Zhang, Ran Lu, Jinglin Wang, Jiwen Yang, Jianqiao Yan, Yuyao Yang, Wenqi Wang, Zhiyong Liu, and Bingyang Wang. Distill any depth: Distillation creates a stronger monocular depth estimator. *arXiv preprint arXiv:2502.19204*, 2025.
- [6] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.
- [7] Jin Han Lee, Myung-Kyu Han, Dong Wook Ko, and Il Hong Suh. From big to small: Multi-scale local planar guidance for monocular depth estimation. *arXiv preprint arXiv:1907.10326*, 2019.
- [8] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Mahmoud Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, Gabriel Synnaeve, Hu Xu, Herve Jegou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. Dinov2: Learning robust visual features without supervision. *arXiv preprint arXiv:2304.07193*, 2023.
- [9] René Ranftl, Alexey Bochkovskiy, and Vladlen Koltun. Vision transformers for dense prediction. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 12179–12188, 2021.
- [10] René Ranftl, Katrin Lasinger, David Hafner, Konrad Schindler, and Vladlen Koltun. Towards robust monocular depth estimation: Mixing datasets for zero-shot cross-dataset transfer. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 44(3):1623–1637, 2022.
- [11] Nathan Silberman, Derek Hoiem, Pushmeet Kohli, and Rob Fergus. Indoor segmentation and support inference from rgbd images. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pages 746–760. Springer, 2012.
- [12] Lihe Yang, Bingyi Kang, Zilong Huang, Xiaogang Xu, Jiashi Feng, and Hengshuang Zhao. Depth anything: Unleashing the power of large-scale unlabeled data. In *Proceedings of*

108 *the IEEE/CVF Conference on Computer Vision and Pattern*
109 *Recognition (CVPR)*, 2024.
110 [13] Lihe Yang, Bingyi Kang, Zilong Huang, Zhen Zhao, Xiao-
111 gang Xu, Jiashi Feng, and Hengshuang Zhao. Depth any-
112 thing v2. *arXiv preprint arXiv:2406.09414*, 2024.
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